

Full-frame terminalization and posterior-covariance bracket boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-27 · this version issued 2026-07-27 · v1.0

1 Purpose and scope

R-096 reduced every rational predictable baseline, after the complete R-077 fresh-root cancellation, to

$$\mathfrak{P}_{k,R} = \mathcal{U}_{R,k} + \tilde{\mathcal{P}}_{R,k}^{\text{pay}} + \frac{1}{2} c_k^T B_{1,k} c_k. \quad (1.1)$$

Estimating this expression separately at each root spends the same prefix energy and terminal sextic reserve repeatedly. R-097 instead recombines all complete Gram rows before applying an inequality.

There are four exact advances. First, a heat-lifted Gram row is a backward heat martingale, so its moving predictable baselines telescope in signed expectation. Second, the single terminal endpoint has one matrix Schur completion at the previously singular boundary parameter $\theta = 0$. Third, conditioning on the complete control and then on the terminal Gram coefficient reduces every remaining signed term to

$$\mathfrak{C}_{\text{post}} = J_B + \mathbb{E}_{\mathcal{F}}[B : (V_B - \Gamma)], \quad (1.2)$$

where $J_B \geq 0$ is an exact Schur compensation. Fourth, a predictable Wick pairing terminalizes into one aggregate coefficient and admits a derivative-free cross-Doob decomposition.

Equation (1.2) is a reduction, not a lower bound. Automatic positivity is false, and predictability alone does not control the aggregate low-Hermite norm. The remaining production target is one weighted lower form for the complete bracket, with J_B , the low endpoint, Cameron–Martin payment, trace, q/r means, and the R-063 forest each used once. The standalone Cartan root budget R-085 (4.11), rational shifted-Hessian form R-085 (6.5), complete H_N , REG, progressive/revisit H_A , OVERLAP_{src}, Nelson, removals, a measure theorem, and Sector A remain open. Tier stays T4.

All identities below are finite-cutoff statements at fixed positive floor. The R-077 regular mutually orthogonal strict-past no-revisit whole-shell class and the fixed R-096 support collar remain the production hypotheses for the rational embedding.

2 Heat-lifted Gram coordinates

Fix a terminal root J . Let

$$X_r = \sum_{s \leq r} g_s, \quad D_i X_r = \sum_{s \leq r} d_{s,i}, \quad (2.1)$$

where the shell increments are independent and jointly Gaussian. Write

$$\Delta \Sigma_r = \text{Cov}(g_r), \quad \Delta \Gamma_{r,i} = \text{Cov}(d_{r,i}), \quad \Gamma_{r,i} = \sum_{s \leq r} \Delta \Gamma_{s,i}. \quad (2.2)$$

The common real-even regulator gives, pointwise,

$$\text{Cov}(g_r(x), d_{r,i}(x)) = 0. \quad (2.3)$$

Thus value and derivative increments at one point are independent.

Let $B : \mathbb{R}^N \rightarrow \text{Sym}_N^+$ be a smooth Gram coefficient of polynomial growth. The two choices used later are the rational Gram

$$B_R(z) = 4c_1 g(z) \otimes g(z) \quad (2.4)$$

and the complete production Gram

$$B_{\text{fr}}(z) = \sum_{\alpha} M_{\alpha}(z) Q_{\Pi} M_{\alpha}(z)^T. \quad (2.5)$$

For future value covariance

$$\Sigma_{r+1:J} = \sum_{s=r+1}^J \Delta \Sigma_s,$$

put

$$\overline{B}_r^{(J)} = P_{\Sigma_{r+1:J}} B. \quad (2.6)$$

For an $\mathcal{F}_{r-1}$ -measurable shift H , define

$$\mathcal{W}_{r,B}^{(J)}(H) = \frac{1}{2} \sum_i \int \overline{B}_r^{(J)}(X_r + H) : ((D_i X_r + D_i H)^{\otimes 2} - \Gamma_{r,i}) dx. \quad (2.7)$$

At $r = J$ this is the literal terminal covariance-normal Gram energy.

3 Complete Gram-row heat telescope

Theorem 3.1 (complete Gram-row heat martingale)

For every admissible $H \in \mathcal{F}_{r-1}$,

$$\boxed{\mathbb{E}[\mathcal{W}_{r,B}^{(J)}(H) \mid \mathcal{F}_{r-1}] = \mathcal{W}_{r-1,B}^{(J)}(H)}. \quad (3.1)$$

Proof. Condition on $\mathcal{F}_{r-1}$ and at one spatial point put

$$z = X_{r-1} + H, \quad y_i = D_i X_{r-1} + D_i H.$$

By (2.3), g_r and $d_{r,i}$ are independent. Hence

$$\begin{aligned} \mathbb{E}_{r-1} \left[\overline{B}_r^{(J)}(z + g_r) : ((y_i + d_{r,i})^{\otimes 2} - \Gamma_{r,i}) \right] \\ = \mathbb{E}_{g_r} \overline{B}_r^{(J)}(z + g_r) : (y_i^{\otimes 2} - \Gamma_{r-1,i}). \end{aligned} \quad (3.2)$$

Heat-semigroup composition gives

$$\mathbb{E}_{g_r} \overline{B}_r^{(J)}(z + g_r) = P_{\Delta \Sigma_r} \overline{B}_r^{(J)}(z) = \overline{B}_{r-1}^{(J)}(z). \quad (3.3)$$

Sum in i , integrate in x , and use finite-cutoff Fubini. $\square$

Let

$$a_k \in \mathcal{F}_{k-1}, \quad A^{(k)} = A^0 + \sum_{j_0 \leq s \leq k} a_s. \quad (3.4)$$

Define

$$\mathfrak{P}_{k,B} = \mathcal{W}_{k-1,B}^{(J)}(A^{(k)}) - \mathcal{W}_{k-1,B}^{(J)}(A^{(k-1)}) \quad (3.5)$$

and

$$\mathfrak{M}_{k,B} = \mathcal{W}_{k,B}^{(J)}(A^{(k)}) - \mathcal{W}_{k-1,B}^{(J)}(A^{(k)}). \quad (3.6)$$

Then

$$\mathbb{E}[\mathfrak{M}_{k,B} \mid \mathcal{F}_{k-1}] = 0 \quad (3.7)$$

and pathwise

$$\mathfrak{P}_{k,B} = \mathcal{W}_{k,B}^{(J)}(A^{(k)}) - \mathcal{W}_{k-1,B}^{(J)}(A^{(k-1)}) - \mathfrak{M}_{k,B}. \quad (3.8)$$

The first two terms telescope, so

$$\mathbb{E} \sum_{k=j_0}^J \mathfrak{P}_{k,B} = \mathbb{E} \left[\mathcal{W}_{J,B}^{(J)}(A^*) - \mathcal{W}_{j_0-1,B}^{(J)}(A^0) \right]. \quad (3.9)$$

This is an expectation identity. The correction (3.6) remains pathwise.

R-077 cancellation must be applied to the complete fresh-root packet before (3.1). The theorem concerns a reassembled covariance-normal Gram row; it does not center an individual chaos, Taylor family, trace fragment, or rational projection.

4 Rational baseline terminalization

For $B = B_R$, R-096 identifies the exact complete rational baseline as

$$\mathfrak{P}_{k,R} = \mathcal{U}_{R,k} + \tilde{\mathcal{P}}_{R,k}^{\text{pay}} + \frac{1}{2} c_k^T B_{1,k} c_k. \quad (4.1)$$

The tilde includes the fixed payable R-096 collar. Equations (3.9) and (4.1) imply

$$\begin{aligned} \mathbb{E} \sum_k \left[\mathcal{U}_{R,k} + \tilde{\mathcal{P}}_{R,k}^{\text{pay}} + \frac{1}{2} c_k^T B_{1,k} c_k \right] \\ = \mathbb{E} \left[\mathcal{W}_{J,R}^{(J)}(A^*) - \mathcal{W}_{j_0-1,R}^{(J)}(A^0) \right]. \end{aligned} \quad (4.2)$$

All five R-085 Taylor families, every payable orientation, the collar, and the endpoint squares have been recombined before any inequality. The R-063 forest already reconstructs the coefficient–Wick product inside (4.2); adding another forest would duplicate its owner.

5 One terminal Schur completion

At the terminal scale put

$$z_0 = X_J + A^0, \quad a = A^* - A^0, \quad G_i = D_i z_0, \quad c_i = D_i a, \quad (5.1)$$

and

$$B_0 = B(z_0), \quad B_1 = B(z_0 + a), \quad Q_i = G_i^{\otimes 2} - \Gamma_{J,i}. \quad (5.2)$$

Let $R = R^T \succ 0$ be one allocated matrix payment and define

$$\mathcal{A} = B_1 + 2R, \quad \Theta_R(B_1) = B_1 - B_1 \mathcal{A}^{-1} B_1. \quad (5.3)$$

Theorem 5.1 (terminal $\theta = 0$ Schur identity)

At finite cutoff, pathwise,

$$\begin{aligned}
& \mathcal{W}_{J,B}^{(J)}(A^*) - \mathcal{W}_{J,B}^{(J)}(A^0) + \sum_i \int c_i^T R c_i \, dx \\
&= \frac{1}{2} \sum_i \int \|c_i + \mathcal{A}^{-1} B_1 G_i\|_{\mathcal{A}}^2 \, dx \\
&+ \frac{1}{2} \sum_i \int (G_i^T \Theta_R(B_1) G_i - B_1 : \Gamma_{J,i}) \, dx - \frac{1}{2} \sum_i \int B_0 : Q_i \, dx.
\end{aligned} \tag{5.4}$$

Restoring the once-only baseline gives

$$\begin{aligned}
& \mathcal{W}_{J,B}^{(J)}(A^*) + \sum_i \int c_i^T R c_i \, dx \\
&= \frac{1}{2} \sum_i \int \|c_i + \mathcal{A}^{-1} B_1 G_i\|_{\mathcal{A}}^2 \, dx \\
&+ \frac{1}{2} \sum_i \int (G_i^T \Theta_R(B_1) G_i - B_1 : \Gamma_{J,i}) \, dx.
\end{aligned} \tag{5.5}$$

Proof. The endpoint expansion is

$$\begin{aligned}
& \mathcal{W}_{J,B}^{(J)}(A^*) - \mathcal{W}_{J,B}^{(J)}(A^0) \\
&= \frac{1}{2} \sum_i \int (B_1 - B_0) : Q_i \, dx + \sum_i \int G_i^T B_1 c_i \, dx + \frac{1}{2} \sum_i \int c_i^T B_1 c_i \, dx.
\end{aligned} \tag{5.6}$$

Expanding the first square in (5.4) produces

$$\frac{1}{2} c_i^T (B_1 + 2R) c_i + G_i^T B_1 c_i + \frac{1}{2} G_i^T B_1 \mathcal{A}^{-1} B_1 G_i. \tag{5.7}$$

The Θ_R term cancels the last expression and leaves $G_i^T B_1 G_i/2$. The trace and baseline then recover (5.6). $\square$

This is the R-095 perspective identity at $\theta = 0$. Since $2R \succeq 0$, $\Theta_R(B_1) \succeq 0$. Nevertheless neither positive term in (5.5) may be discarded before the trace is treated: both conditional variances are load-bearing.

6 Posterior-covariance normal form

Let $\mathcal{F}$ make the complete terminal shift, c_i , and R measurable. Put

$$\bar{B} = \mathbb{E}_{\mathcal{F}} B_1, \quad q_i = \mathbb{E}_{\mathcal{F}}(B_1 G_i), \quad \bar{\mathcal{A}} = \bar{B} + 2R. \tag{6.1}$$

Condition further on B_1 and define

$$\mu_{B,i} = \mathbb{E}[G_i \mid \mathcal{F} \vee \sigma(B_1)], \quad V_{B,i} = \text{Cov}(G_i \mid \mathcal{F} \vee \sigma(B_1)). \tag{6.2}$$

Then $q_i = \mathbb{E}_{\mathcal{F}}(B_1 \mu_{B,i})$. Define

$$J_{B,i} = \mathbb{E}_{\mathcal{F}}[\mu_{B,i}^T B_1 \mu_{B,i}] - q_i^T \bar{\mathcal{A}}^{-1} q_i. \tag{6.3}$$

Theorem 6.1 (posterior-covariance bracket identity)

For every i , $J_{B,i} \geq 0$, and

$$\begin{aligned} \mathcal{S}_{0,B}(c) &:= \frac{1}{2} \sum_i \int \mathbb{E}_{\mathcal{F}}[B_1 : (G_i^{\otimes 2} - \Gamma_{J,i})], dx \\ &\quad + \sum_i \int c_i^T q_i dx + \frac{1}{2} \sum_i \int c_i^T \bar{\mathcal{A}} c_i dx \\ &= \frac{1}{2} \sum_i \int \|c_i + \bar{\mathcal{A}}^{-1} q_i\|_{\bar{\mathcal{A}}}^2 dx + \frac{1}{2} \mathfrak{C}_{\text{post}}, \end{aligned} \tag{6.4}$$

where

$$\mathfrak{C}_{\text{post}} = \sum_i \int \{J_{B,i} + \mathbb{E}_{\mathcal{F}}[B_1 : (V_{B,i} - \Gamma_{J,i})]\} dx. \tag{6.5}$$

Proof. For fixed i, x , the block matrix

$$\begin{pmatrix} B_1 & B_1 \mu_{B,i} \\ \mu_{B,i}^T B_1 & \mu_{B,i}^T B_1 \mu_{B,i} \end{pmatrix} \tag{6.6}$$

is positive semidefinite. Take $\mathbb{E}_{\mathcal{F}}$, add $2R$ to its upper-left block, and take the Schur complement to obtain $J_{B,i} \geq 0$. Furthermore,

$$\mathbb{E}_{\mathcal{F}}[B_1 : G_i^{\otimes 2}] = \mathbb{E}_{\mathcal{F}}[B_1 : V_{B,i}] + \mathbb{E}_{\mathcal{F}}[\mu_{B,i}^T B_1 \mu_{B,i}]. \tag{6.7}$$

Completing the c_i square gives

$$\frac{1}{2} c_i^T \bar{\mathcal{A}} c_i + c_i^T q_i = \frac{1}{2} \|c_i + \bar{\mathcal{A}}^{-1} q_i\|_{\bar{\mathcal{A}}}^2 - \frac{1}{2} q_i^T \bar{\mathcal{A}}^{-1} q_i. \tag{6.8}$$

Insert (6.7)–(6.8) into (6.4). □

The second term in (6.5) is sign-indefinite. The nonnegative J_B must remain coupled to it; estimating the raw covariance deficit alone can lose an almost exact cancellation.

7 Mean and forest ownership

Let

$$C_1 = \Theta_R(B_1)^{1/2}, \quad r_i = \mathbb{E}_{\mathcal{F}}(C_1 G_i), \tag{7.1}$$

and define

$$\mathfrak{D}_R = \sum_i \int \left\{ \mathbb{E}_{\mathcal{F}}[B_1 : (G_i^{\otimes 2} - \Gamma_{J,i})] - q_i^T \bar{\mathcal{A}}^{-1} q_i - |r_i|^2 \right\} dx. \tag{7.2}$$

Equations (6.3)–(6.7) give

$$\mathfrak{C}_{\text{post}} = \mathfrak{D}_R + \sum_i \int |r_i|^2 dx. \tag{7.3}$$

Thus the once-only owners are:

1. the raw Wick term owns the trace and complete R-063 forest;
2. q_i belongs to the mixed current and completed c_i square;
3. r_i is restored by terminal perspective energy;

4. $J_{B,i}$ is coefficient-reveal perspective variance;
5. $B_0 : Q_i/2$ is the low endpoint owner in (5.4); and
6. the fixed R-096 collar stays in $\tilde{\mathcal{P}}_{R,k}^{\text{pay}}$.

No item may be appended twice. Under the extra R-096 conditional Gaussian hypothesis, the raw Wick owner sees only coefficient Hermite ranks zero, one, and two. Theorem 6.1 itself differentiates no adapted selector.

8 Derivative-free predictable terminalization

Write $G_j = \sum_{s \leq j} g_s$, let $Q_j = G_j^{\otimes 2} - \Gamma_j$ be the quadratic Wick martingale of the revealed roots, and let B_k be predictable with deterministic spatial map T_k . Define

$$F_J = \sum_k T_k^* B_k. \quad (8.1)$$

For $0 \leq j \leq J$, set $F_j = \mathbb{E}[F_J \mid \mathcal{F}_j]$. For $1 \leq j \leq J$, set $d_j F = F_j - F_{j-1}$ and $d_j Q = Q_j - Q_{j-1}$. Then the tower property gives the exact terminalization

$$\boxed{\sum_k \mathbb{E} \langle B_k, T_k Q_{k-1} \rangle = \mathbb{E} \langle F_J, Q_J \rangle.} \quad (8.2)$$

For every starting level ℓ ,

$$\mathbb{E} \langle F_J, Q_J \rangle = \mathbb{E} \langle F_\ell, Q_\ell \rangle + \sum_{j > \ell} \mathbb{E} \langle d_j F, d_j Q \rangle. \quad (8.3)$$

Writing $\Delta \Gamma_j = \text{Cov}(g_j)$, the tensor identity is

$$d_j Q = G_{j-1} \otimes g_j + g_j \otimes G_{j-1} + (g_j^{\otimes 2} - \Delta \Gamma_j). \quad (8.4)$$

After contraction with a symmetric scalar unit-variance coordinate this is the shorthand $d_j Q = 2G_{j-1}g_j + H_2(g_j)$. Hence only the first two conditional Hermite ranks of $d_j F$ pair with $d_j Q$. No Malliavin derivative of a selector is introduced.

This identity does not itself provide a root-uniform estimate. If one old Gaussian root ξ_1 is reused through

$$B_k = H_2(\xi_1) \quad (k > 1), \quad (8.5)$$

then N uses give pairing $2N$ but aggregate norm

$$\left\| \sum_{k=1}^N B_k \right\|_2^2 = 2N^2. \quad (8.6)$$

Thus predictability alone is insufficient. A production proof needs a weighted low-Hermite/Doob norm with genuine spatial or root gain, or a direct signed cancellation.

9 Rational and full-frame consequences

Corollary 9.1 (rational terminal form)

Under the R-077/R-096 hypotheses,

$$\begin{aligned}
& \mathbb{E} \sum_k \left[\mathcal{U}_{R,k} + \tilde{\mathcal{P}}_{R,k}^{\text{pay}} + \frac{1}{2} c_k^T B_{1,k} c_k \right] + \mathbb{E} \mathcal{W}_{j_0-1,R}^{(J)}(A^0) \\
& + \mathbb{E} \sum_i \int c_i^T R c_i \, dx \\
& = \frac{1}{2} \mathbb{E} \sum_i \int \|c_i + \bar{\mathcal{A}}^{-1} q_i\|_{\bar{\mathcal{A}}}^2 \, dx + \frac{1}{2} \mathbb{E} \mathfrak{C}_{\text{post}}(B_R, G; R \mid \mathcal{F}).
\end{aligned} \tag{9.1}$$

Consequently R-096 (8.3) is reduced to one lower form for the last bracket, the low endpoint, and the single allocated matrix payment. No per-root adapted coefficient norm is needed.

Corollary 9.2 (full-frame integrated successor)

For $B = B_{\text{fr}}$, the same argument gives the sufficient new target

$$\begin{aligned}
& \frac{1}{2} \mathbb{E} \mathfrak{C}_{\text{post}}(B_{\text{fr}}, G; R \mid \mathcal{F}) - \mathbb{E} \mathcal{W}_{j_0-1,\text{fr}}^{(J)}(A^0) \\
& - \mathbb{E} \sum_i \int c_i^T R c_i \, dx - \mathbb{E} \Delta \mathcal{R}_{\leq} \\
& \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta,\zeta}.
\end{aligned} \tag{9.2}$$

Here X and Y are the globally owned Cameron–Martin and terminal sextic budgets in the R-066/R-079 ledger, while $\Delta \mathcal{R}_{\leq}$ is exactly the low-frequency remainder of R-079 (4.2), carried through R-096 (2.8); none is newly estimated here. Conditional on separate once-only inequalities paying those inherited owners with remaining η/ζ slack, the new analytic input sufficient for (9.2) is

$$\boxed{\mathbb{E} \mathfrak{C}_{\text{post}}(B_{\text{fr}}, G; R \mid \mathcal{F}) \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta,\zeta}.} \tag{9.3}$$

The bracket is nonlinear in B_{fr} , so cross-row Schur compensation is retained. This creates a possible joint route for the rational mechanism R-085 (6.5) and the nonlinear full-frame mechanism whose separate Cartan architecture requires (4.10)–(4.11).

Equation (9.3) does not prove the standalone Cartan spatial estimate (4.10), its one-use root budget (4.11), or the rational-only bound (6.5). The spatial half (4.10) remains the earlier R-087 result; (4.11) remains open. R-097 is an alternative integrated architecture, not a retroactive proof of those standalone statements.

10 Exact method boundaries

Two independent exact fixtures prevent false closure.

First, take a Rademacher root $\xi \in \{-1, 1\}$ and

$$B(\xi) = 2 + \xi, \quad R = 10^{-2}. \tag{10.1}$$

Then $Q = \xi^2 - 1 = 0$ atomwise, while

$$\bar{B} = 2, \quad q = \mathbb{E}(B\xi) = 1. \tag{10.2}$$

The paid scalar block

$$\mathcal{S}_R(c) = qc + \frac{1}{2}(\bar{B} + 2R)c^2 \quad (10.3)$$

has exact minimum

$$\inf_c \mathcal{S}_R(c) = -\frac{1}{2(2 + 2R)} < 0. \quad (10.4)$$

The transformed mean $r = \mathbb{E}[\Theta_R(B)^{1/2}\xi]$ is nonzero and restores its ownership exactly; it does not make (10.4) positive. Direct sums show that bounded coefficients plus predictability alone cannot yield a root-count-uniform abstract constant.

Second, let $g \sim N(0, 1)$ and

$$B(g) = (g + 2)^2 + 1, \quad Q = H_2(g). \quad (10.5)$$

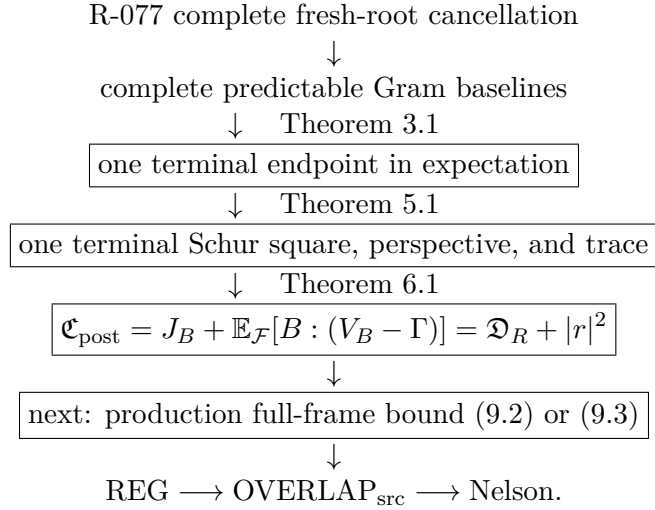
The complete forest is derived rather than truncated:

$$BQ = H_4 + 4H_3 + 10H_2 + 8H_1 + 2H_0. \quad (10.6)$$

Here $\bar{B} = 6$, $q = 4$, and $\mathbb{E}(BQ) = 2$, so at $R = 10^{-2}$ the terminal Schur determinant is negative. Every rank zero through four is a genuine owner. Omitting or duplicating any component breaks reconstruction.

These fixtures are finite abstract coefficient/direct-sum method obstructions. They are not counterexamples realized in the production torus geometry, and they neither prove nor refute (9.3).

11 Proof and failure map



The route verdicts are:

- complete Gram heat telescope: proved at finite cutoff;
- pathwise baseline telescope: false because (3.6) remains;
- terminal $\theta = 0$ Schur identity: proved;
- posterior-covariance normal form: proved;
- automatic posterior-bracket positivity: false;
- predictability-only aggregate control: false;
- separate per-baseline Taylor payment: retired as primary route;
- production full-frame posterior form: open and required; and
- standalone rational (6.5) and Cartan (4.11): still open.

12 Devil’s-advocate self-test

1. **Objection: rational projection may precede R-077 cancellation.**

UPHELD as false. Theorem 3.1 applies only after the complete packet has been canceled and reassembled as a Gram row. No individual fresh rational, trace, or forest fragment is centered.

2. **Objection: predictable baselines telescope pathwise.**

UPHELD as false. Equation (3.8) contains the diagonal heat martingale $\mathfrak{M}_{k,B}$. Only its signed conditional expectation vanishes.

3. **Objection: $\Theta_R(B_1) \succeq 0$ makes the endpoint nonnegative.**

UPHELD as false. The Wick trace remains signed. Conditional variances from both positive terms in (5.5) are load-bearing.

4. **Objection: the R-063 forest pays q or r .**

UPHELD as an ownership error. The forest reconstructs raw $B : Q$. The mean q belongs to the completed control square, while r belongs to terminal perspective energy.

5. **Objection: value-derivative independence implies $V_B = \Gamma$ after conditioning on an adapted selector.**

UPHELD as false. Unconditional heat independence does not preserve derivative covariance after coefficient or selector revelation.

6. **Objection: J_B may be discarded and the covariance deficit bounded alone.**

UPHELD as a potentially fatal loss. The positive Schur term can cancel almost all of the negative posterior covariance term. Their sum is the exact target.

7. **Objection: $R = B_1$ may be chosen without bookkeeping.**

UPHELD as false. The algebra allows random R , but the quadratic cost must come from an actual once-only Cameron–Martin reserve. Moving or same-root choices also carry the R-092/R-095 defects.

8. **Objection: (9.3) has already proved R-085 (4.11) and (6.5).**

UPHELD as an overclaim. Equation (9.3) is still open and is a joint full-frame successor. It need not imply either standalone statement.

9. **Objection: finite-cutoff identities give adapted continuum reconstruction.**

UPHELD as false. R-063 supplies deterministic H^2 translation, not a control-independent norm for an adapted terminal coefficient.

10. **Objection: the finite fixtures disprove the production theorem.**

DISMISSED. They refute automatic algebraic positivity and an abstract predictability-only constant. Production spatial/root structure may still prove (9.3).

13 Verification and authority map

The primary executable checks the heat telescope, five noncommuting matrix Schur fixtures, posterior conditional identities, cross-Doob decomposition, low-Hermite projection, repeated-root accumulation, complete Gaussian forest, and deliberate mutations. The independent executable imports only the Python

standard library and recomputes exact Fraction-valued Schur, telescope, forest, posterior covariance, direct-sum, and perspective certificates.

Run from the repository root:

```
python codes/foundations/
a13_classii_global_gram_terminalization_
covariance_deficit_reduction_verify.py
```

The authority chain is R-063 for the complete forest, R-066 for backward heat current identities, R-077 for complete fresh-root cancellation, R-079 for the full safe frame, R-085/R-086 for Cartan and rational coordinates, R-087 for the Cartan spatial half, R-093/R-095 for perspective and conditional-mean debts, and R-096 for the adapted-prefix frontier.

14 Result footer

Result ID: A13-CLASSII-GLOBAL-GRAM-TERMINALIZATION-
COVARIANCE-DEFICIT-REDUCTION

Ledger: R-097.

Title: Full-frame terminalization and posterior-covariance bracket
boundary.

Precise statement:

Complete heat-lifted Gram rows obey a backward-heat martingale identity. Predictable Gram baselines telescope in expectation. The terminal endpoint plus one matrix payment has an exact $\theta=0$ Schur form. Conditional completion isolates $C_{\text{post}} = J_B + E[B:(V_B - \Gamma)] = D_R + |r|^2$, $J_B \geq 0$.

Scope: Finite cutoff; positive floor; common real-even regulator; strict-past controls for which every displayed polynomial Gram and derivative product is integrable. Rational support collapse uses the R-077 regular no-revisit whole-shell class and fixed collar.

Dependencies:

R-063; R-066; R-077; R-079; R-085; R-086; R-087; R-093;
R-095; R-096.

Evidence grade:

ANALYTIC, EXACT, EXECUTED, REDUCTION, METHOD BOUNDARY.

Reproduction command:

```
python codes/foundations/a13_classii_global_gram_
terminalization_covariance_deficit_reduction_verify.py
```

Expected output:

Primary 91/91; independent 69/69; integrated 189/189;
aggregate 349/349; exit zero.

Falsification gate:

Any heat, Schur, covariance, Doob, forest, fixture, mutation,
authority, hash, PDF, count, ownership, or T4 scope check fails.

Tier before / after:

T4 / T4; no promotion.

No-overclaim statement:

No production posterior-covariance lower form, Cartan one-use (4.11), rational-only (6.5), complete H_N , REG, H_A , OVERLAP_src, Nelson, removals, measure, T5--T7, or Sector-A closure is proved.

Next required action:

Prove the full-frame posterior bracket bound (9.2) or (9.3), retaining J_B , low endpoint, CM payment, subtraction, q/r , trace, and forest exactly once.